

Name: \_\_\_\_\_

Date: \_\_\_\_\_

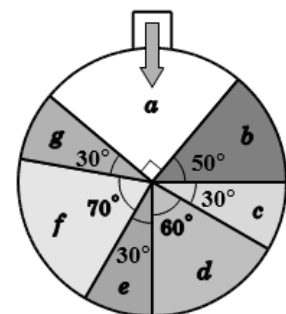
- A letter is chosen randomly from the word 'UNIVERSITY'. Find the probability that is a vowel.
  - $\frac{1}{5}$
  - $\frac{3}{10}$
  - $\frac{2}{5}$
  - $\frac{1}{2}$

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- Samuel can choose to go to work by bus, taxi or MTR. Suppose he randomly chooses a mean of transport in two working days. Find the probability that Samuel chooses the same mean of transport in these two working days.
  - $\frac{2}{3}$
  - $\frac{1}{3}$
  - $\frac{1}{4}$
  - $\frac{3}{5}$

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- There are 20 basketball and 35 tennis ball in a box. Nick repeats the action of drawing a ball from the box at random and putting it back into the box. Find the expected number of times of getting a tennis ball if he repeats the action for 385 times.
  - 20
  - 35
  - 140
  - 245

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- The figure shows a circular fortune wheel in a lucky draw. Dylan turns the wheel once. If the pointer points at the region 'c', 'e' or 'g', a prize will be given. Find the probability that he gets a prize.
 

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- $\frac{1}{4}$
- $\frac{1}{6}$
- $\frac{1}{10}$
- $\frac{1}{12}$

5. A bag contains five \$1.4 stamps, five \$0.2 stamps and ten \$0.1 stamps. A stamp is drawn at random from the bag. Find the expected face value of the stamp.
- A. \$0.35  
B. \$0.4  
C. \$0.45  
D. \$0.5
6. There are 24 bottles of water and  $x$  bottles of tea on a table. If a bottle of drink is drawn at random, the probability of drawing a bottle of tea is  $\frac{3}{5}$ . Find the value of  $x$ .
- A. 36  
B. 38  
C. 40  
D. 42
7.  $6\blacktriangle$  is a 2-digit number, where  $\blacktriangle$  is an integer from 0 to 9 inclusive. Find the probability that the 2-digit number is
- (a) greater than 60 (3 marks)  
(b) divisible by 5 (2 marks)
8. There are 2000 candidates in an examination. If one of the candidates is chosen randomly, the probability of choosing a female candidate is  $\frac{11}{20}$ . Find the number of male candidates in the examination. (3 marks)

9. Two fair dice are thrown at the same time. By using tabulation, find the probability that
- (a) one of the two numbers obtained is a multiple of 3. (5 marks)
  - (b) the sum of the two number is less than 9. (2 marks)

\*10. Winnie's purse contains two \$2 coins, one \$5 coin and one \$10 coin. On a flag day, Winnie takes out two coins randomly from her purse at the same time for donation.

- (a) Find the probability that the donation amount is between \$5 and \$13. (5 marks)
- (b) Find the expected donation amount. (4 marks)